05 Internetworking

05.01-05.02
ARP and ICMP

Types of Addresses in Internet

Media Access Control (MAC) addresses in the network access layer
- *Associated w/ network interface card (NIC)*
- *48 bits or 64 bits*

IP addresses for the network layer
- *32 bits for IPv4, and 128 bits for IPv6*
- *E.g., 123.4.56.7*

IP addresses + ports for the transport layer
- *E.g., 123.4.56.7:80*

Domain names for the application/human layer
- *E.g., www.google.com*

IP And MAC working together

•IP addresses are chosen by the local system administrator to suit the local network

•Ethernet addresses are built into the interface hardware by the manufacturer

•The two addresses bear absolutely no relationship to one another (as we would expect from the layering principles)

Translation of Addresses

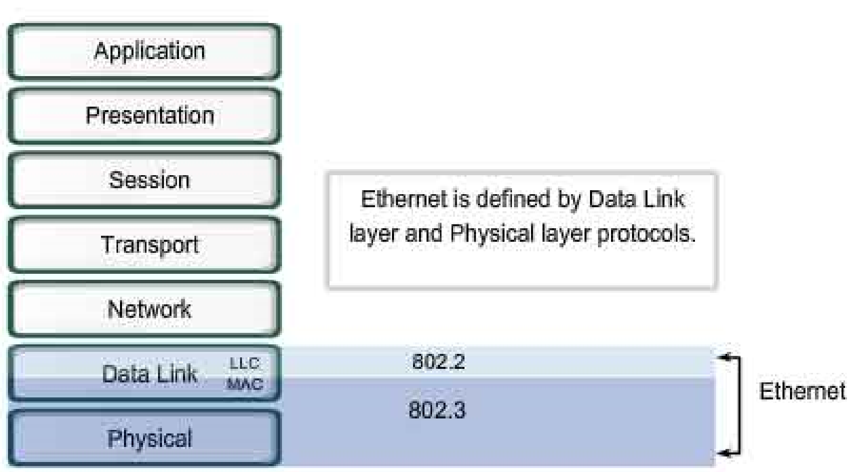
- Translation between IP addresses and MAC addresses
  - •Address Resolution Protocol (ARP) for IPv4
  - •Neighbor Discovery Protocol (NDP) for IPv6

•Translation between IP addresses and domain names (Domain Name System (DNS))

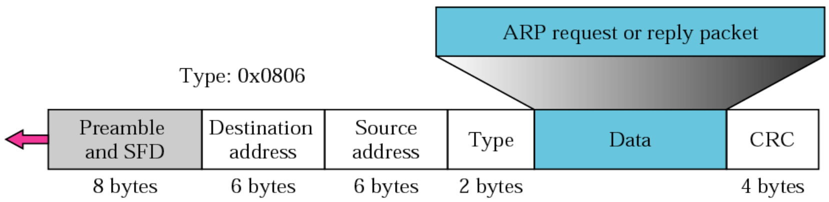
ARP Basics

- Usually considered to be a part of the link layer
- The Data Link layer has (e.g., 6 byte Ethernet) addresses, while the network layer has independent (4 byte) IP addresses
- Primarily used to translate IP addresses to Ethernet MAC addresses
  - The device drive for Ethernet NIC needs to do this to send a packet
- Also used for IP over other LAN technologies, e.g., FDDI, or IEEE 802.11

Ethernet

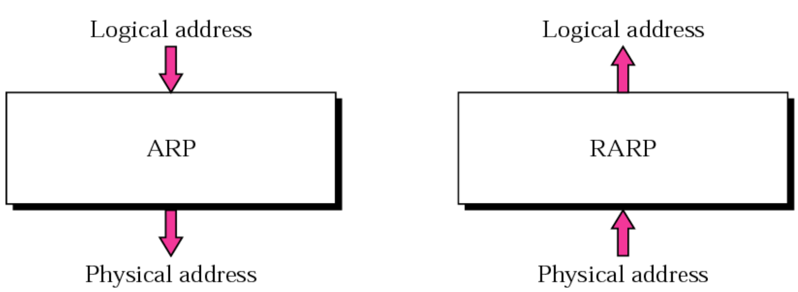


ARP in Ethernet packet



- *The ARP packet is encapsulated within an Ethernet packet.*
- *Note: Type field for Ethernet is x0806*

ARP and RARP



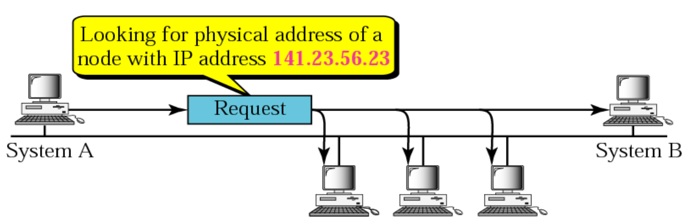
What is ARP used for?

•*Suppose want to send a packet over (say) an Ethernet.*

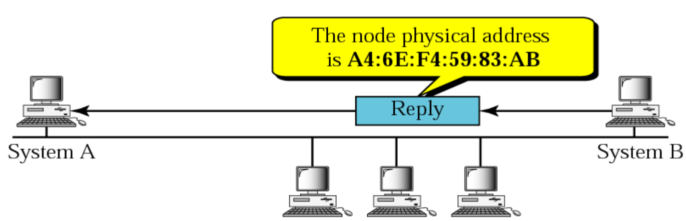
• *We only know the destination's IP address to build the Ethernet frame we have to know the Ethernet address that the destination has.*

•*This is what ARP does: Find the hardware address corresponding to an IP address*

ARP sample

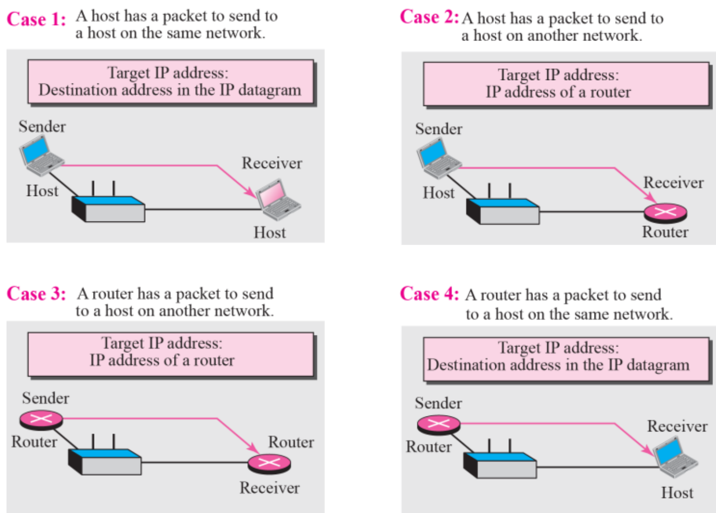


a. ARP request is broadcast

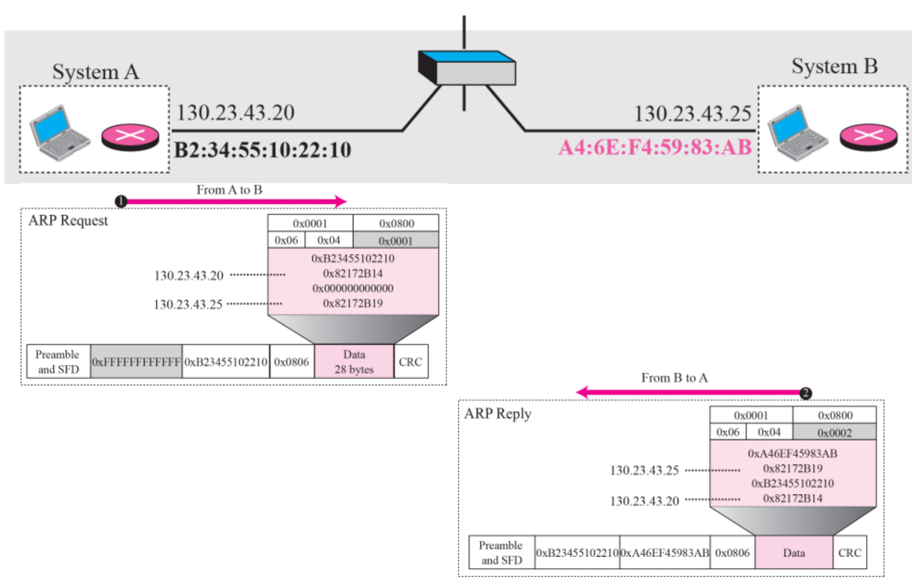


b. ARP reply is unicast

Four cases using ARP



ARP process



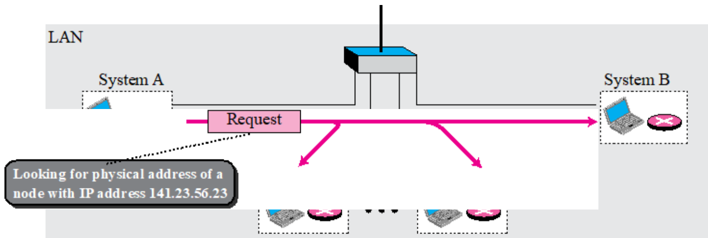
ARP Walkthrough Pt 1

1. ARP broadcasts an ARP Request packet that contains the target IP address in an Ethernet frame with destination address ff:ff:ff:ff:ff:ff (and source its own Ethernet address)
2. All hosts on the local network read the frame
3. The target host recognises the request for its IP address

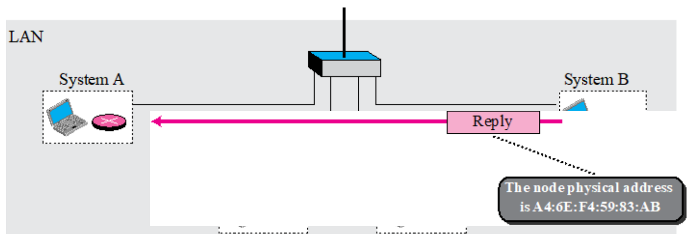
ARP Walkthrough Pt 2

1. The target sends an ARP Reply packet containing its own Ethernet address (the other hosts need do nothing)
2. It knows the source's Ethernet address as read from the request packet
3. The source gets the reply and reads out the target's Ethernet address
4. It can now use that Ethernet address to send IP packets

Still ARP



a. ARP request is multicast



b. ARP reply is unicast

ARP Cache

- For every outgoing packet sending ARP request and waiting for responses is inefficient
- Requires more bandwidth
- Consumes Time
- ARP cache maintained at each node
- Size limit = 512 entries (timer)

The Cache Table

If ARP just resolved an IP address, chances are a few moments later someone is going to ask to resolve the same IP address

- When ARP returns a MAC address, it is placed in a cache. When the next request comes in for the same IP address, look first in the cache

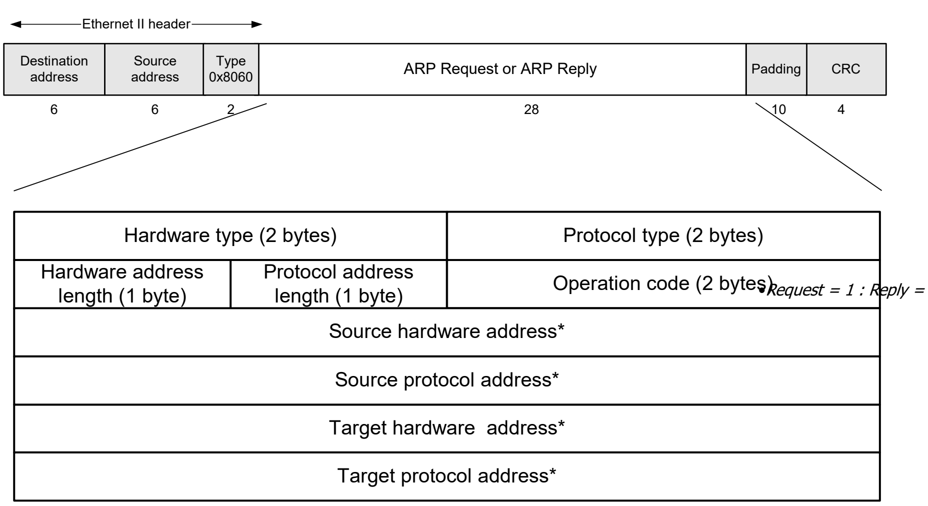
Cache Table

- Each host maintains a table of IP to MAC addresses
- Message types:
  - ARP request
  - ARP reply
  - ARP announcement

ARP Cache Problems

- Cache space may be limited
- Hosts move or change IP addresses
- Solution?
- Drop (invalidate) cache entries after “a while” (20 minutes is normal)

ARP Packet Format

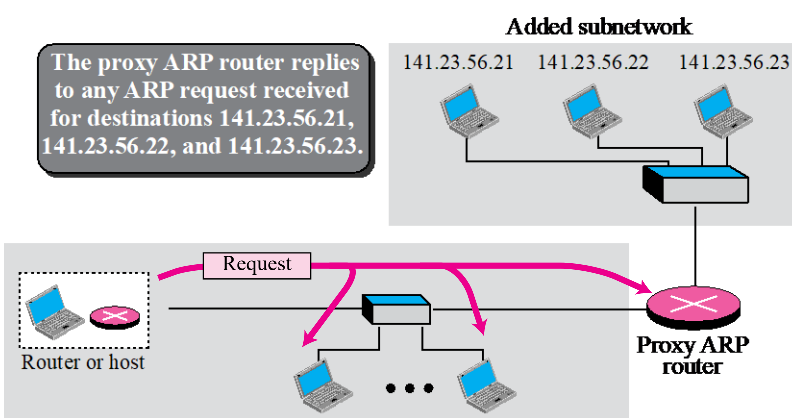


* Note: The length of the address fields is determined by the corresponding address length fields

•Proxy Arp

• Host or router responds to ARP Request that arrives from one of its connected networks for a host that is on another of its connected networks

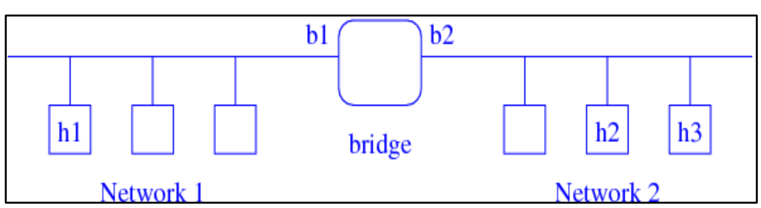
Proxy ARP



ARP Command

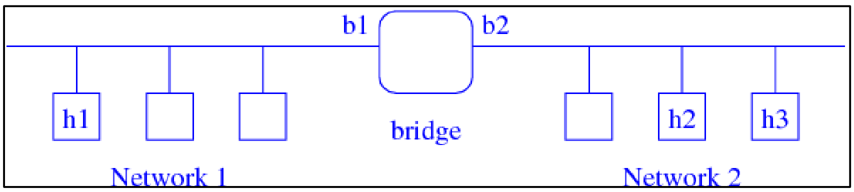
- To display table
  arp –a
- To enter manually (Static Entry)
  arp -s 192.168.1.2 00-FE-FE-FE-FE-FE
- To delete entry
  arp –d 192.168.1.2

ARP Bridging



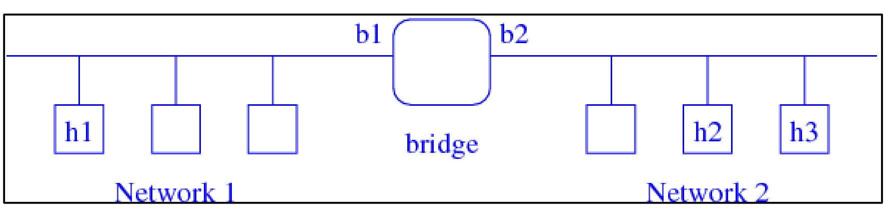
- A bridge is a host with two interfaces, one on each network
- If host h1 wishes to send to host h2 it must determine its hardware address

ARP Bridging



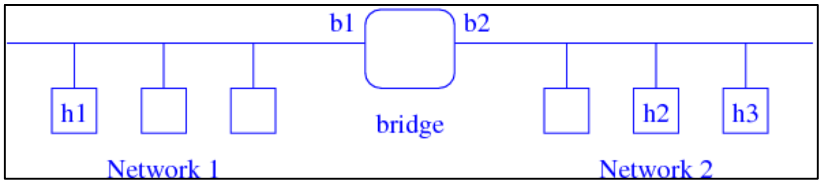
- So h1 sends an ARP broadcast for h2
- The bridge sees this request and responds on behalf of h2 (a proxy ARP), but it supplies its own hardware address b1

ARP Bridging



- Now h1 sends data to what it thinks is h2, but is actually the bridge
- The bridge reads the packet, sees it is destined for h2 (by its IP address) and forwards it to the other network where h2 can read it

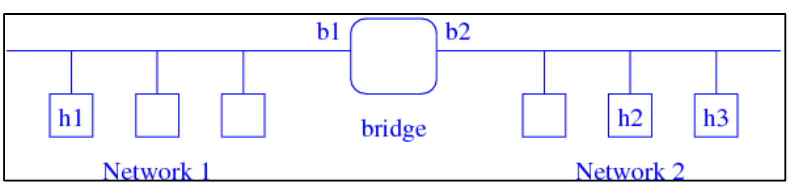
ARP Bridging



- In either case the packet goes to the bridge, which forwards it to h1, again rewriting the frame addresses appropriately
- This is all transparent to h1 and h2 who believe they are on the same network

Computer Networks
05.01 05.02 ARP and ICMP
applied computer science
urbino worldwide campus

ARP Bridging



- This is sometimes called transparent bridging
- If h1 is communicating with both h2 and h3 its cache will show then to have the same hardware address b1: this is not a problem

ARP Bridging

- ARP bridging is fine for joining a pair of small networks, but less so for larger collections of networks
- IEEE 802.1d Ethernet Bridging standard addresses this, dealing with the cases of multiple routes between hosts

ARP Spoofing (ARP Poisoning)

Send fake or 'spoofed', ARP messages to an Ethernet LAN.

- To have other machines associate IP addresses with the attacker’s MAC

Defenses

- Static ARP table
- DHCP snooping (use access control to ensure that hosts only use the IP addresses assigned to them, and that only authorized DHCP servers are accessible).
- Detection: Arpwatch (sending email when updates occur),

Legitimate use

- Redirect a user to a registration page before allow usage of the network

Internet Control Message Protocol (ICMP)

RFC 792

- **What is its purpose?**
  - The purpose of control messages is to provide feedback about problems in the communication environment.
  - ICMP messages typically report errors in the datagram processing.
  - Both a gateway or a host can use the ICMP.
    - A few uses for ICMP messages could be:
      - when a datagram cannot reach its destination,
      - when the gateway does not have the buffering capacity to forward a datagram,
      - when the gateway can direct the host to send traffic on a shorter route.

ICMP Message Format

- Type
  - the type of service being provided. There's a specific type number for each error or informational message sent.
- Code
  - the error code provides further information on the message type. It tells what was the possible cause to the problem.
- Checksum
  - the 16-bit one's complement of the one's complement sum of the ICMP message starting with the ICMP type. Used to find problems on the ICMP message ONLY.
- Parameters
  - used in some specific ICMP messages to exchange other information such as pointers, identifiers, and sequence numbers.
- Additional Info
  - Used on error report messages, includes the header plus additional octets from the datagram that caused the problem.

•*ICMP Message:*

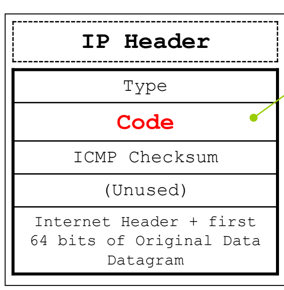
IPv4 Header	
Type	8 bytes
Code	8 bytes
ICMP Checksum	16 bytes
Parameters, if any	Var.
Additional Info	Var.

ICMP Messages: Summary

Type	Message	Query/ Error	Description
0	Echo Reply	Query	Response given when a sender issues an Echo Request
3	Destination Unreachable	Error	The destination host cannot be reached
4	Source Quench	Error	Source host is sending datagrams too fast
5	Redirect	Error	Used by a router to notify a host of a better route
8	Echo Request	Query	Used to check whether communication with a host is possible
11	Time Exceeded	Error	Time-To-Live (TTL) has reached 0
12	Parameter Problem	Error	There was a problem with the IP Header
13	Timestamp Request	Query	Used for sampling the delay characteristics of the network
14	Timestamp Reply	Query	Used for reporting how long it took for a Timestamp Request to reach a host
15	Information Request	Query	Used by a host to discover the address of the network it is on
16	Information Reply	Query	Used for replying an Information Request message with the network address

Destination Unreachable Message

Type 3



Code	Reason
0	net unreachable
1	host unreachable
2	protocol unreachable
3	port unreachable
4	fragmentation needed and DF flag is set
5	source route failed

- Possible scenarios this message may be sent:

1. When, according to the gateway's routing tables, the network specified in the destination field of a datagram is unreachable.
2. When the IP module cannot deliver the datagram because the indicated protocol module or process port is not active.
3. When the datagram must be fragmented to be forwarded by a gateway yet the Don't Fragment (DF) flag is on.

Source Quench Message

Type 4

IP Header
Type
Code
ICMP Checksum
(Unused)
Internet Header + first 64 bits of Original Data Datagram

Code	Reason
0	(no special meaning)

- Possible scenarios this message may be sent:

1. When a gateway does not have the buffer space needed to queue the datagrams for output to the next network on the route to the destination network.

Redirect Message

Type 5

IP Header
Type
Code
ICMP Checksum
Gateway Internet Address
Internet Header + first 64 bits of Original Data Datagram

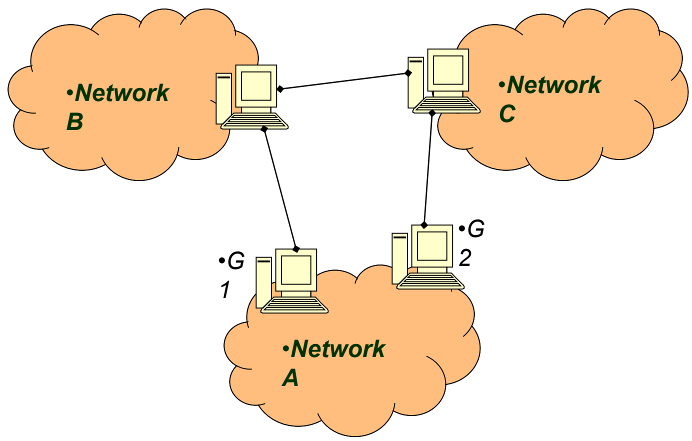
Code	Meaning
0	redirect datagrams for the Network
1	redirect datagrams for the Host
2	redirect datagrams for the Type of Service and Network
3	redirect datagrams for the Type of Service and Host

•Indicates the address of the Gateway to which traffic for the network specified in the internet destination network field of the original datagram's data should be sent

- Possible scenarios this message may be sent:

1. When a gateway, after checking its routing tables, finds out that there is a shorter path to the destination host passing through another route.

Redirect Message: Example



Echo Request/Echo Reply Message

Type 8 / Type 0

IP Header
Type
Code
ICMP Checksum
Identifier | Seq.Num.
Internet Header + first 64 bits of Original Data Datagram

Type	Meaning
8	echo (request) message
0	echo reply message

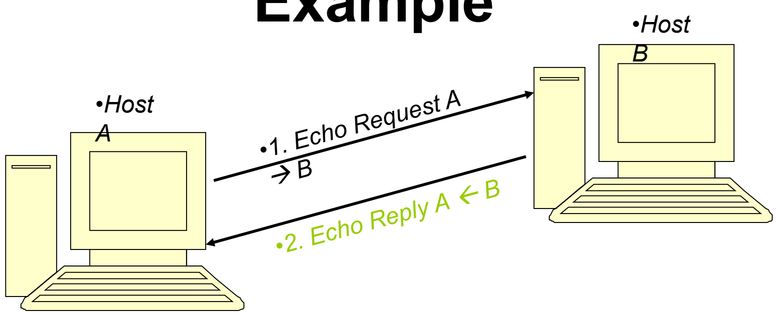
Code	Meaning
0	(no special meaning)

•The Identifier and the Sequence Number fields are used to aid in matching echoes and replies. For example, the identifier might be used like a port in TCP or UDP to identify a session, and the sequence number might be incremented on each echo request sent.

- Possible scenarios this message may be sent:

1. When a host or gateway wants to check if communication with a host is possible.

Echo Request/Reply: Example



1. Host A sends an Echo Request Message containing the source address of Host A and destination address of Host B, using Identifier "5350" and Sequence Number "40"
2. Host B replies with an Echo Reply Message with source and destination addresses from the original message reversed, repeating the same Identifier "5350" and Sequence Number "40"

Computer Networks
05.01 05.02 ARP and ICMP
applied computer science
urbino worldwide campus

Time Exceeded Message

Type 11

IP Header
Type
Code
ICMP Checksum
(Unused)
Internet Header + first 64 bits of Original Data Datagram

Code	Reason
0	time to live exceeded in transit
1	fragment reassembly time exceeded

- Possible scenarios this message may be sent:

1. When the gateway processing a datagram finds that the Time-To-Live field is 0.
2. When a host reassembling a fragmented datagram cannot complete the reassembly due to missing fragments within its time limit.

Parameter Problem Message

Type 12

IP Header
Type
Code
ICMP Checksum
Pointer | (Unused)
Internet Header + first 64 bits of Original Data Datagram

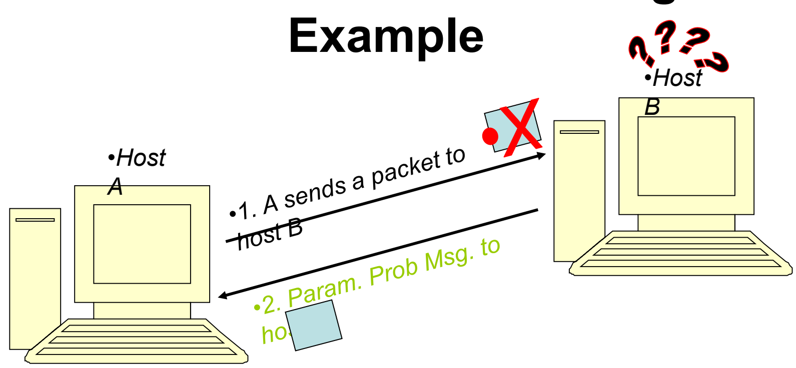
Code	Reason
0	pointer indicates the error

- *When the code is 0, the pointer indicates the octet where an error was detected.*

- Possible scenarios this message may be sent:

1. When a gateway or host finds a problem with the header parameters so that it cannot complete processing the datagram.

Parameter Problem Message: Example



1. Host A sends a packet to Host B, but host B cannot complete processing the datagram because it found errors in arguments sent in an option.
2. Host B replies with a Parameter Problem Message back to Host A, with Pointer = 20 (which may indicate a problem with Type of Service of the first option, for instance), and the first octets from the offending packet sent by A.

Timestamp or Timestamp Reply Message

Type 13 / Type 14

IP Header
Type
Code
ICMP Checksum
Identifier | Seq.Num.
Originate Timestamp*
Receive Timestamp*
Transmit Timestamp*

Type	Meaning
13	timestamp (request) message
14	timestamp reply message

Code	Meaning
0	(no special meaning)

• *The Identifier and the Sequence Number fields are used to aid in matching timestamp and replies.*

•*A timestamp is 32 bits of milliseconds since midnight.

- Possible scenarios this message may be sent:

1. When a host or gateway wants to check the delay characteristics of the network.

Information Request / Information Reply Message

Type 15 / Type 16

IP Header
Type
Code
ICMP Checksum
Identifier | Seq.Num.

Type	Meaning
15	information request message
16	information reply message

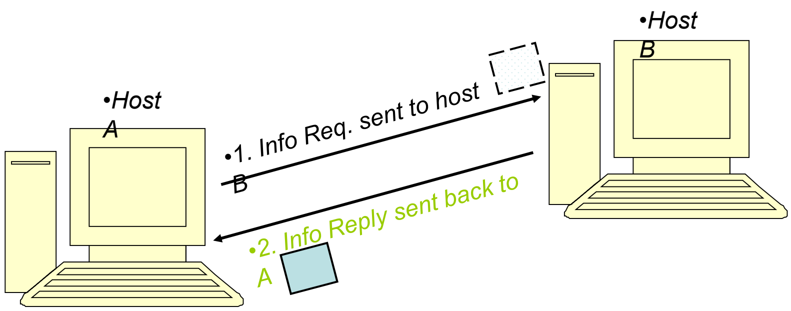
Code	Meaning
0	(no special meaning)

• *The requesting host sends the message with the network portion of the source and destination IP address field set to zero.*

- Possible scenarios this message may be sent:

1. When a host wants to discover the address of the network it is on.

Information Request/Reply: Example



1. Host A sends a packet to Host B, with the source network in the IP header source and destination address fields zero (which means "this" network).
2. Host B (the replying IP module) should send the reply with the addresses fully specified. This message is a way for a host to find out the number of the network it is on.

Internet Control Message Protocol version 6

RFC 2463

- Same as ICMP for IPv4, but with a number of changes.
- The IPv6 version of ICMP includes a pseudo-header in its checksum computation.
- The reason for the change is to protect ICMP from misdelivery or corruption of those fields of the IPv6 header on which it depends, which, unlike IPv4, are not covered by an internet-layer checksum.
- The Next Header field in the pseudo-header for ICMP contains the value 58, which identifies the IPv6 version of ICMP.
- ICMPv6 is an integral part of IPv6 and MUST be fully implemented by every IPv6 node.
- ICMPv6 is used by IPv6 nodes to report errors encountered in processing packets, and to perform other internet-layer functions, such as diagnostics (ICMPv6 "ping").

ICMPv6 Message Format

- ICMPv6 messages are grouped into two classes:
  1. Error Messages
  2. Informational Messages
- <u>Error messages</u> are identified as such by having a zero in the high-order bit of their message Type field values.
- Thus, error messages have message types ranging from 0 to 127, and informational messages ranging from 128 to 255.

IPv6 Header
Type
Code
ICMPv6 Checksum
Message Body

ICMPv6: Message Source Address Determination

- A node that sends an ICMPv6 message has to determine <u>both the Source and Destination IPv6 Addresses</u> in the IPv6 header <u>before calculating the checksum</u>. If the node has more than one unicast address, it must choose the Source Address of the message as follows:

1. If the message is a response to a message sent to one of the node's unicast addresses, the Source Address of the reply must be **<u>that same address</u>**.
2. If the message is a response to a message sent to a multicast or anycast group in which the node is a member, the Source Address of the reply must be **<u>a unicast address belonging to the interface on which the multicast or anycast packet was received</u>.**
3. If the message is a response to a message sent to an address that does NOT belong to the node, the Source Address should be **<u>that unicast address belonging to the node that will be most helpful in diagnosing the error</u>**.
4. Otherwise, the node's routing table must be examined to determine which interface will be used to transmit the message to its destination, and a unicast address belonging to that interface must be used as the Source Address of the message.

ICMPv6: Message Processing Rules

Directly from the RFC 1122

- Implementations MUST observe the following rules when processing ICMPv6 messages:
  1. If an ICMPv6 error message of unknown type is received, it MUST be passed to the upper layer.
  2. If an ICMPv6 informational message of unknown type is received, it MUST be silently discarded.
  3. Every ICMPv6 error message includes as much of the IPv6 offending packet (the packet that caused the error) as will fit without making the error message packet exceed the minimum IPv6 MTU.
  4. In those cases where the internet-layer protocol is required to pass an ICMPv6 error message to the upper-layer process, the upper-layer protocol type is extracted from the original packet (contained in the body of the ICMPv6 error message) and used to select the appropriate upper-layer process to handle the error.
  5. An ICMPv6 error message MUST NOT be sent as a result of receiving (1) an ICMPv6 error message; (2) a packet destined to an IPv6 multicast address; (3) a packet sent as a link-layer multicast; (4) a packet sent as a link-layer broadcast; (5) a packet whose source address does not uniquely identify a single node.
  6. Finally, in order to limit the bandwidth and forwarding costs incurred sending ICMPv6 error messages, an IPv6 node MUST limit the rate of ICMPv6 error messages it sends.

ICMPv6 Messages: Summary

Type	Message	Query/ Error	Description
1	Destination Unreachable	Error	The destination host cannot be reached
2	Packet Too Big	Error	The packet cannot be forwarded because it is larger than the MTU of the outgoing link
3	Time Exceeded	Error	Hop Limit Value has reached 0
4	Parameter Problem	Error	Cannot complete processing the packet because there is a problem with a field in the IPv6 header or extension headers such that it cannot complete processing the packet.
128	Echo Request	Query	Used for diagnostic purposes
129	Echo Reply	Query	Message sent in response to an Echo Request Message.

Destination Unreachable Message

ICMPv6 :: Type 1

IP Header
Type
Code
Checksum
(Unused)
As much of invoking packet as will fit without the ICMPv6 packet exceeding the min. IPv6 MTU.

Code	Reason
0	no route to destination
1	communication to destination administratively prohibited
2	(not assigned)
3	address unreachable
4	port unreachable

•Unused for all code values. It must be initialized to zero by the sender and ignored by the receiver.

- Possible scenarios this message may be sent:

1. When, according to the gateway's routing tables, the network specified in the destination field of a datagram is unreachable.
2. When the IP module cannot deliver the datagram because the indicated protocol module or process port is not active.
3. When there exists a "firewall filter" due to administrative prohibition.

Packet Too Big Message

ICMPv6 :: Type 2

IP Header
Type
Code
Checksum
MTU
As much of invoking packet as will fit without the ICMPv6 packet exceeding the min. IPv6 MTU.

Code	Reason
0	(set to 0 by the sender and ignored by the receiver)

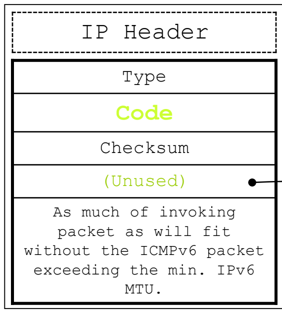
•The Maximum Transmission Unit of the next-hop link

- *Possible scenarios this message may be sent:*

1. *When a router cannot forward because it is larger than the MTU of the outgoing link.*

Time Exceeded Message

ICMPv6 :: Type 3



Code	Reason
0	hop limit exceeded in transit
1	fragment reassembly time exceeded

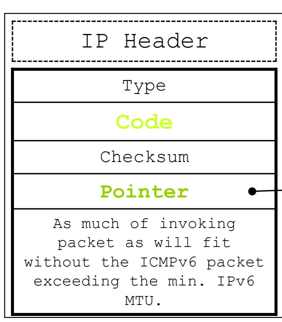
•Unused for all code values. It must be initialized to zero by the sender and ignored by the receiver.

- *Possible scenarios this message may be sent:*

1. *When a router receives a packet with a Hop Limit of zero*
2. *When a router decrements a packet's Hop Limit to zero.*

Parameter Problem Message

ICMPv6 :: Type 4



Code	Reason
0	erroneous header field encountered
1	unrecognized Next Header type encountered
2	unrecognized IPv6 option encountered

•Identifies the octet offset within the invoking packet where the error was detected.

- Possible scenarios this message may be sent:

1. When an IPv6 node processing a packet finds a problem with a field in the IPv6 header or extension headers such that it cannot complete processing the packet.

Echo Request Message

ICMPv6 :: Type 128

IP Header
Type
Code
Checksum
Identifier | Seq.Num.
Data*…

•* Zero or more octets of arbitrary data.

Type	Meaning
0	no meaning

Code	Meaning
0	(no special meaning)

•The Identifier and the Sequence Number fields are used to aid in matching echoes and replies.

- Possible scenarios this message may be sent:

1. When a node wants to check if communication with a host is possible. Used for diagnostic purposes.

Computer Networks
05.01 05.02 ARP and ICMP
applied computer science
urbino worldwide campus

Echo Reply Message

ICMPv6 :: Type 129

IP Header
Type
Code
Checksum
Identifier | Seq.Num.
Data*…

• Zero or more octets of arbitrary data.*

Type	Meaning
0	no meaning

Code	Meaning
0	(no special meaning)

•The Identifier, Sequence Number, and Data fields from the invoking Echo Request message

- *Possible scenarios this message may be sent:*

1. *When a node gets an Echo Request Message.*

References

- RFC 792 found at http://www.ietf.org/rfc/rfc0792
- RFC 2463 found at http://www.ietf.org/rfc/rfc2463